

BABU BANARASI DAS UNIVERSITY

LUCKNOW

Session : 2026-2027

SCHOOL OF COMPUTER APPLICATIONS

**ASSIGNMENT
OF**

NoSQL and DBaaS 101

(BCADSN23202)

Submitted By:

VIKAS SHARMA

BCA DS & AI

BCADS27

Roll No. 1250258503

Submitted To:

Mr. Harendra Singh

MONGODB STUDENT DATABASE PROJECT

Complete MongoDB Shell (mongosh) Operations

Database: studentProjectDB | Collection: studentRecords

PROJECT INFORMATION

Database Name: studentProjectDB

Collection Name: studentRecords

Total initial records: 51 (Roll No. 201-251)

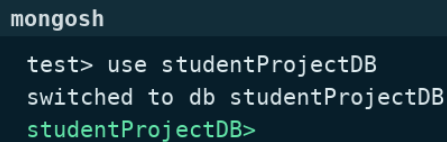
Operations covered: Insert One, Insert Many, Update One, Update Many, Delete One, Delete Many, Query Operations and Verification.

1. CREATE / SELECT DATABASE

Question: How do you create or select the database for the MongoDB student project?

```
use studentProjectDB;
```

SCREENSHOT / OUTPUT



```
mongosh
test> use studentProjectDB
switched to db studentProjectDB
studentProjectDB>
```

2. INSERT ONE

Question: Write a MongoDB command to insert one student record into the studentRecords collection.

```
db.studentRecords.insertOne({
  rollNo: 201,
  name: "Ananya",
  age: 19,
  department: "CSE",
  marks: 91
});
```

SCREENSHOT / OUTPUT

mongosh

```
studentProjectDB> db.studentRecords.insertOne({
  rollNo: 201, name: 'Ananya', age: 19,
  department: 'CSE', marks: 91
})
{ acknowledged: true, insertedId: ObjectId('33400c474439b4f2b6d6317e') }
studentProjectDB>
```

3. INSERT MANY

Question: Write a MongoDB command to insert multiple student records (Roll No. 202–251) into the studentRecords collection.

```
db.studentRecords.insertMany([
  { rollNo: 202, name: "Raghav", age: 20, department: "IT", marks: 76 },
  { rollNo: 203, name: "Meera", age: 18, department: "ECE", marks: 84 },
  { rollNo: 204, name: "Arjun", age: 21, department: "ME", marks: 67 },
  { rollNo: 205, name: "Kavya", age: 19, department: "CSE", marks: 88 },
  { rollNo: 206, name: "Naman", age: 20, department: "IT", marks: 79 },
  { rollNo: 207, name: "Ishita", age: 18, department: "ECE", marks: 42 },
  { rollNo: 208, name: "Rohit", age: 21, department: "ME", marks: 94 },
  { rollNo: 209, name: "Aditi", age: 19, department: "CSE", marks: 73 },
  { rollNo: 210, name: "Kunal", age: 20, department: "IT", marks: 86 },
  { rollNo: 211, name: "Sneha", age: 18, department: "ECE", marks: 55 },
  { rollNo: 212, name: "Manav", age: 21, department: "ME", marks: 81 },
  { rollNo: 213, name: "Pooja", age: 19, department: "CSE", marks: 96 },
  { rollNo: 214, name: "Varun", age: 20, department: "IT", marks: 68 },
  { rollNo: 215, name: "Rahul", age: 18, department: "ECE", marks: 87 },
  { rollNo: 216, name: "Simran", age: 21, department: "ME", marks: 74 },
  { rollNo: 217, name: "Neha", age: 19, department: "CSE", marks: 61 },
  { rollNo: 218, name: "Aditya", age: 20, department: "IT", marks: 99 },
  { rollNo: 219, name: "Tanvi", age: 18, department: "ECE", marks: 78 },
  { rollNo: 220, name: "Vivek", age: 21, department: "ME", marks: 89 },
  { rollNo: 221, name: "Sakshi", age: 19, department: "CSE", marks: 35 },
  { rollNo: 222, name: "Harsh", age: 20, department: "IT", marks: 72 },
  { rollNo: 223, name: "Priya", age: 18, department: "ECE", marks: 93 },
  { rollNo: 224, name: "Aman", age: 21, department: "ME", marks: 65 },
  { rollNo: 225, name: "Riya", age: 19, department: "CSE", marks: 82 },
  { rollNo: 226, name: "Mohit", age: 20, department: "IT", marks: 58 },
  { rollNo: 227, name: "Nidhi", age: 18, department: "ECE", marks: 71 },
  { rollNo: 228, name: "Yash", age: 21, department: "ME", marks: 97 },
  { rollNo: 229, name: "Isha", age: 19, department: "CSE", marks: 45 },
  { rollNo: 230, name: "Deepak", age: 20, department: "IT", marks: 83 },
  { rollNo: 231, name: "Muskan", age: 18, department: "ECE", marks: 64 },
  { rollNo: 232, name: "Akash", age: 21, department: "ME", marks: 77 },
  { rollNo: 233, name: "Ritu", age: 19, department: "CSE", marks: 90 },
  { rollNo: 234, name: "Saurabh", age: 20, department: "IT", marks: 37 },
  { rollNo: 235, name: "Komal", age: 18, department: "ECE", marks: 69 },
```

3. INSERT MANY - CONTINUED

```
{ rollNo: 229, name: "Isha", age: 19, department: "CSE", marks: 45 },
{ rollNo: 230, name: "Deepak", age: 20, department: "IT", marks: 83 },
{ rollNo: 231, name: "Muskan", age: 18, department: "ECE", marks: 64 },
{ rollNo: 232, name: "Akash", age: 21, department: "ME", marks: 77 },
{ rollNo: 233, name: "Ritu", age: 19, department: "CSE", marks: 90 },
{ rollNo: 234, name: "Saurabh", age: 20, department: "IT", marks: 37 },
{ rollNo: 235, name: "Komal", age: 18, department: "ECE", marks: 69 },
{ rollNo: 236, name: "Rajat", age: 21, department: "ME", marks: 85 },
{ rollNo: 237, name: "Payal", age: 19, department: "CSE", marks: 57 },
{ rollNo: 238, name: "Varsha", age: 20, department: "IT", marks: 92 },
{ rollNo: 239, name: "Nikhil", age: 18, department: "ECE", marks: 48 },
{ rollNo: 240, name: "Shreya", age: 21, department: "ME", marks: 80 },
{ rollNo: 241, name: "Divya", age: 19, department: "CSE", marks: 75 },
{ rollNo: 242, name: "Manish", age: 20, department: "IT", marks: 66 },
{ rollNo: 243, name: "Pallavi", age: 18, department: "ECE", marks: 39 },
{ rollNo: 244, name: "Rohan", age: 21, department: "ME", marks: 91 },
{ rollNo: 245, name: "Bhavna", age: 19, department: "CSE", marks: 33 },
{ rollNo: 246, name: "Tarun", age: 20, department: "IT", marks: 88 },
{ rollNo: 247, name: "Kriti", age: 18, department: "ECE", marks: 81 },
{ rollNo: 248, name: "Gaurav", age: 21, department: "ME", marks: 34 },
{ rollNo: 249, name: "Alisha", age: 19, department: "CSE", marks: 95 },
{ rollNo: 250, name: "Abhishek", age: 20, department: "IT", marks: 36 },
{ rollNo: 251, name: "Sonali", age: 18, department: "ECE", marks: 43 }
]);
```

SCREENSHOT / OUTPUT

SCREENSHOT / OUTPUT

mongosh

```
studentProjectDB> db.studentRecords.insertMany([
  { rollNo: 202, name: 'Raghav', age: 20, department: 'IT', marks: 76 },
  { rollNo: 203, name: 'Meera', age: 18, department: 'ECE', marks: 84 },
  ...
  { rollNo: 251, name: 'Sonali', age: 18, department: 'ECE', marks: 43 }
])
{ acknowledged: true, insertedIds: { '0': ObjectId('...'), ... } }
studentProjectDB>
```

4. UPDATE ONE

Question: How do you update the marks of the student with Roll No. 201 from 91 to 95?

```
db.studentRecords.updateOne(  
  { rollNo: 201 },  
  { $set: { marks: 95 } }  
);
```

SCREENSHOT / OUTPUT

mongosh

```
studentProjectDB> db.studentRecords.updateOne({ rollNo: 201 },  
  { $set: { marks: 95 } })  
{ acknowledged: true, matchedCount: 1, modifiedCount: 1 }  
studentProjectDB>
```

5. UPDATE MANY

Question: How do you add the status "Active" to all students belonging to the CSE department?

```
db.studentRecords.updateMany(  
  { department: "CSE" },  
  { $set: { status: "Active" } }  
);
```

SCREENSHOT / OUTPUT

mongosh

```
studentProjectDB> db.studentRecords.updateMany(
  { department: 'CSE' }, { $set: { status: 'Active' } })
{ acknowledged: true, matchedCount: 13, modifiedCount: 13 }
studentProjectDB>
```

6. DELETE ONE

Question: How do you delete the student record having Roll No. 251?

```
db.studentRecords.deleteOne({ rollNo: 251 });
```

SCREENSHOT / OUTPUT

mongosh

```
studentProjectDB> db.studentRecords.deleteOne({ rollNo: 251 })
{ acknowledged: true, deletedCount: 1 }
studentProjectDB>
```

7. DELETE MANY

Question: How do you delete all student records having marks less than 40?

```
db.studentRecords.deleteMany({
  marks: { $lt: 40 }
});
```

SCREENSHOT / OUTPUT

mongosh

```
studentProjectDB> db.studentRecords.deleteMany({ marks: { $lt: 40 } })
{ acknowledged: true, deletedCount: 6 }
studentProjectDB>
```

8. QUERY OPERATIONS

A. MARKS GREATER THAN 80

Question: How do you find all students whose marks are greater than 80? (\$gt)

```
db.studentRecords.find({
  marks: { $gt: 80 }
});
```

SCREENSHOT / OUTPUT

SCREENSHOT / OUTPUT

mongosh

```
studentProjectDB> db.studentRecords.find(marks > 80)
```

201	Ananya	19	CSE	95	Active
203	Meera	18	ECE	84	
205	Kavya	19	CSE	88	Active
208	Rohit	21	ME	94	
210	Kunal	20	IT	86	
212	Manav	21	ME	81	
213	Pooja	19	CSE	96	Active

Type ! to get more (14 more records)

B. MARKS LESS THAN 80

Question: How do you find all students whose marks are less than 80? (\$lt)

```
db.studentRecords.find({
  marks: { $lt: 80 }
});
```

SCREENSHOT / OUTPUT

mongosh

```
studentProjectDB> db.studentRecords.find(marks < 80)
```

202	Raghav	20	IT	76	
204	Arjun	21	ME	67	
206	Naman	20	IT	79	
207	Ishita	18	ECE	42	
209	Aditi	19	CSE	73	Active
211	Sneha	18	ECE	55	
214	Varun	20	IT	68	

Turn title for name (15 name marks)

C. MARKS EQUAL TO 85

Question: How do you find all students whose marks are exactly 85? (\$eq)

```
db.studentRecords.find({
  marks: { $eq: 85 }
});
```

C. MARKS EQUAL TO 85

SCREENSHOT / OUTPUT

mongosh

```
studentProjectDB> db.studentRecords.find(marks = 85)
236  Rajat      21  ME    85
studentProjectDB>
```

D. MARKS GREATER THAN OR EQUAL TO 80

Question: How do you find all students whose marks are greater than or equal to 80? (\$gte)

```
db.studentRecords.find({
  marks: { $gte: 80 }
});
```

SCREENSHOT / OUTPUT

```
mongosh
studentProjectDB> db.studentRecords.find(marks >= 80)
201  Ananya      19  CSE  95  Active
203  Meera       18  ECE  84
205  Kavya       19  CSE  88  Active
208  Rohit       21  ME   94
210  Kunal       20  IT   86
212  Manav       21  ME   81
213  Pooja       19  CSE  96  Active
Type 'it' for more (15 more records)
```

E. MARKS NOT EQUAL TO 85

Question: How do you find all students whose marks are not equal to 85? (\$ne)

```
db.studentRecords.find({  
  marks: { $ne: 85 }  
});
```

SCREENSHOT / OUTPUT

SCREENSHOT / OUTPUT

mongosh

```
studentProjectDB> db.studentRecords.find(marks != 85)
```

201	Ananya	19	CSE	95	Active
202	Raghav	20	IT	76	
203	Meera	18	ECE	84	
204	Arjun	21	ME	67	
205	Kavya	19	CSE	88	Active
206	Naman	20	IT	79	
207	Ishita	18	ECE	42	

Type !t to see more (26 more records)

F. MARKS GREATER THAN 70 AND AGE LESS THAN 23

Question: How do you find students whose marks are greater than 70 AND age is less than 23? (\$and)

```
db.studentRecords.find({
  $and: [
    { marks: { $gt: 70 } },
    { age: { $lt: 23 } }
  ]
});
```

SCREENSHOT / OUTPUT

SCREENSHOT / OUTPUT

mongosh

```
studentProjectDB> db.studentRecords.find(marks > 70 AND age < 23)
```

201	Ananya	19	CSE	95	Active
202	Raghav	20	IT	76	
203	Meera	18	ECE	84	
205	Kavya	19	CSE	88	Active
206	Naman	20	IT	79	
208	Rohit	21	ME	94	
209	Aditi	19	CSE	73	Active

Turn left for more (24 more records)

G. DEPARTMENT IS CSE OR ECE

Question: How do you find students who belong to either CSE OR ECE department? (\$or)

```
db.studentRecords.find({
  $or: [
    { department: "CSE" },
    { department: "ECE" }
  ]
});
```

SCREENSHOT / OUTPUT

mongosh

```
studentProjectDB> db.studentRecords.find(department = CSE OR ECE)
```

201	Ananya	19	CSE	95	Active
203	Meera	18	ECE	84	
205	Kavya	19	CSE	88	Active
207	Ishita	18	ECE	42	
209	Aditi	19	CSE	73	Active
211	Sneha	18	ECE	55	
213	Pooja	19	CSE	96	Active

Turn left for more (15 more records)

H. MARKS NOT GREATER THAN 80

Question: How do you find students whose marks are NOT greater than 80? (\$not)

```
db.studentRecords.find({
  marks: { $not: { $gt: 80 } }
});
```

H. MARKS NOT GREATER THAN 80

SCREENSHOT / OUTPUT

```
mongosh
```

```
studentProjectDB> db.studentRecords.find(NOT (marks > 80))
```

```
202 Raghav      20 IT    76
204 Arjun       21 ME    67
206 Naman       20 IT    79
207 Ishita     18 ECE   42
209 Aditi      19 CSE   73 Active
211 Sneha      18 ECE   55
214 Varun      20 IT    68
Type 'help' for more (16 more records)
```

9. BASIC DISPLAY / VERIFICATION COMMANDS

Show current database

Question: What command is used to show current database?

```
db;
```

9. BASIC DISPLAY / VERIFICATION COMMANDS

Show current database - SCREENSHOT / OUTPUT

```
mongosh
studentProjectDB> db
studentProjectDB
```

Show collections

Question: What command is used to show collections?

```
show collections;
```

SCREENSHOT / OUTPUT

```
mongosh
studentProjectDB> show collections
studentRecords
studentProjectDB>
```

Show all student records

Question: What command is used to show all student records?

```
db.studentRecords.find();
```

SCREENSHOT / OUTPUT

```
mongosh
studentProjectDB> db.studentRecords.find();
{
  _id: ObjectId('33400c474439b4f2b6d6317e'),
  rollNo: 201,
  name: 'Ananya',
  age: 19,
  department: 'CSE',
  marks: 95,
  status: 'Active'
}
```

Show all records in readable format

Question: What command is used to show all records in readable format?

```
db.studentRecords.find().pretty();
```

SCREENSHOT / OUTPUT

mongosh

```
studentProjectDB> db.studentRecords.find().pretty();
```

```
{
  _id: ObjectId('33400c474439b4f2b6d6317e'),
  rollNo: 201,
  name: 'Ananya',
  age: 19,
  department: 'CSE',
  marks: 95,
  status: 'Active'
}
```

Count total documents

Question: What command is used to count total documents?

```
db.studentRecords.countDocuments();
```

SCREENSHOT / OUTPUT

```
mongosh
studentProjectDB> db.studentRecords.countDocuments()
44
studentProjectDB>
```

Count CSE students

Question: What command is used to count CSE students?

```
db.studentRecords.countDocuments({ department: "CSE" });
```

SCREENSHOT / OUTPUT

```
mongosh
studentProjectDB> db.studentRecords.countDocuments({ department: 'CSE' })
11
studentProjectDB>
```

SCREENSHOT / OUTPUT

```
mongosh
studentProjectDB> db.studentRecords.countDocuments({ department: 'CSE' })
11
studentProjectDB>
```

Count ECE students

Question: What command is used to count ECE students?

```
db.studentRecords.countDocuments({ department: "ECE" });
```

SCREENSHOT / OUTPUT

```
mongosh
studentProjectDB> db.studentRecords.countDocuments({ department: 'ECE' })
11
studentProjectDB>
```

Count IT students

Question: What command is used to count IT students?

```
db.studentRecords.countDocuments({ department: "IT" });
```

SCREENSHOT / OUTPUT

```
mongosh
studentProjectDB> db.studentRecords.countDocuments({ department: 'IT' })
11
studentProjectDB>
```

Count ME students

Question: What command is used to count ME students?

```
db.studentRecords.countDocuments({ department: "ME" });
```

SCREENSHOT / OUTPUT

```
mongosh
studentProjectDB> db.studentRecords.countDocuments({ department: 'ME' })
11
studentProjectDB>
```

Find student by Roll No. 201

Question: What command is used to find student by roll no. 201?

```
db.studentRecords.findOne({
  rollNo: 201
});
```

SCREENSHOT / OUTPUT

mongosh

```
studentProjectDB> db.studentRecords.findOne({ rollNo: 201 });
{
  _id: ObjectId('33400c474439b4f2b6d6317e'),
  rollNo: 201,
  name: 'Ananya',
  age: 19,
  department: 'CSE',
  marks: 95,
```

Find all CSE students

Question: What command is used to find all CSE students?

```
db.studentRecords.find({ department: "CSE" });
```

SCREENSHOT / OUTPUT

mongosh

```
studentProjectDB> db.studentRecords.find({ department: 'CSE' });
{
  _id: ObjectId('33400c474439b4f2b6d6317e'),
  rollNo: 201,
  name: 'Ananya',
  age: 19,
  department: 'CSE',
  marks: 95,
  status: 'Active'
}
```

Find all ECE students

Question: What command is used to find all ECE students?

```
db.studentRecords.find({ department: "ECE" });
```

SCREENSHOT / OUTPUT

mongosh

```
studentProjectDB> db.studentRecords.find({ department: 'ECE' });
{
  _id: ObjectId('309ccf017e55eaa09755e1c9'),
  rollNo: 203,
  name: 'Meera',
  age: 18,
  department: 'ECE',
  marks: 84
}
{
  id: ObjectId('617400e49b8ce172ecf7edc0'),
```

Find all IT students

Question: What command is used to find all IT students?

```
db.studentRecords.find({ department: "IT" });
```

SCREENSHOT / OUTPUT

mongosh

```
studentProjectDB> db.studentRecords.find({ department: 'IT' });
{
  _id: ObjectId('c9dfeafbad16577b9a419ec8'),
  rollNo: 202,
  name: 'Raghav',
  age: 20,
  department: 'IT',
  marks: 76
}
{
  id: ObjectId('7d7e5af0caec4142ccbc85e4'),
```

Find all ME students

Question: What command is used to find all ME students?

```
db.studentRecords.find({ department: "ME" });
```

SCREENSHOT / OUTPUT

mongosh

```
studentProjectDB> db.studentRecords.find({ department: 'ME' });
{
  _id: ObjectId('dadccd12f745da815447e0ba'),
  rollNo: 204,
  name: 'Arjun',
  age: 21,
  department: 'ME',
  marks: 67
}
{
  id: ObjectId('51628f5a75a17d1a56e3146f'),
```

STUDENT DATABASE VERIFICATION

The final dataset contains 44 documents after the Delete One and Delete Many operations. Department counts remain balanced at 11 each.

mongosh

```
studentProjectDB> db.studentRecords.find().pretty();
{
  _id: ObjectId('33400c474439b4f2b6d6317e'),
  rollNo: 201,
  name: 'Ananya',
  age: 19,
  department: 'CSE',
  marks: 95,
  status: 'Active'
}
```

OPERATION SUMMARY

- Insert One: rollNo 201
- Insert Many: rollNo 202-251
- Update One: rollNo 201 marks 91 → 95
- Update Many: CSE → status Active
- Delete One: rollNo 251
- Delete Many: marks < 40 → 6 records deleted
- Query operators: \$gt, \$lt, \$eq, \$gte, \$ne, \$and, \$or, \$not
- Final records: 44; CSE 11, ECE 11, IT 11, ME 11

Thank You